

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 1

MH1100 Calculus I

Tutorial 6, Week 7

Your tutor will aim to discuss: Problem 1, 3, 7, 9, and 11.

Problem 1 Prove that f is continuous at a if and only if

$$\lim_{h \rightarrow 0} f(a+h) = f(a).$$

[Solution:] We need to prove two statements: (1) If f is continuous, then $\lim_{h \rightarrow 0} f(a+h) = f(a)$;
(2) If $\lim_{h \rightarrow 0} f(a+h) = f(a)$, then $f(x)$ is continuous at a .

(1) If f is continuous at a , then by Theorem 8 of Handout 5 with $g(h) = a+h$, we have

$$\lim_{h \rightarrow 0} g(h) = \lim_{h \rightarrow 0} (a+h) = a.$$

Thus

$$\lim_{h \rightarrow 0} f(a+h) = \lim_{h \rightarrow 0} f(g(h)) = f(\lim_{h \rightarrow 0} g(h)) = f(\lim_{h \rightarrow 0} (a+h)) = f(a)$$

(2) Let $\epsilon > 0$. Since $\lim_{h \rightarrow 0} f(a+h) = f(a)$, there exists $\delta > 0$ such that if $0 < |h| < \delta$ then $|f(a+h) - f(a)| < \epsilon$. So if $0 < |x-a| < \delta$, then $|f(x) - f(a)| = |f(a+(x-a)) - f(a)| < \epsilon$. Thus, $\lim_{x \rightarrow a} f(x) = f(a)$ and so f is continuous at a .

Problem 2 Explain why some standard theorems about continuous functions imply that the following function is continuous at every point of its domain

$$f(x) = \frac{\tan x}{\sqrt{4-x^2}}.$$

[Solution:] The domain will be the set of points x satisfying the 3 conditions that

(1) x lies in the domain of $\tan x$, which is $\mathbb{R} \setminus \{n\pi + \frac{\pi}{2}\}$ (n is an integer).

(2) x lies in the domain of $\sqrt{4-x^2}$, which is $[-2, 2]$.

(3) The denominator is not zero at that x .

We deduce that the domain of f is $(-2, 2) \setminus \{-\frac{\pi}{2}, \frac{\pi}{2}\}$. Let a be an arbitrary point from the domain $(-2, 2) \setminus \{-\frac{\pi}{2}, \frac{\pi}{2}\}$. For this a we deduce:

(1) The function $4-x^2$ is a polynomial, so, by a standard theorem, it is continuous at every point of $\mathbb{R}$. In particular, it is continuous at a .

- (2) We know that the square root function $\sqrt{x}$ is continuous on the interval $[0, \infty)$. Thus, because $\sqrt{4 - a^2} > 0$, $\sqrt{x}$ is continuous at $4 - a^2$. Thus, by the standard theorem about the composition of two continuous functions being continuous, we deduce that the composition $\sqrt{4 - x^2}$ is continuous at the point a .
- (3) Finally, because $f(x)$ is the quotient of two functions, each of which is continuous at the point a , with the denominator nonzero at a , we deduce from a standard theorem that $f(x)$ is itself continuous at the point a .

Problem 3

- (a) Show that there exists a number $c \in (0, 1)$ where $\sqrt[7]{c} = 1 - c$.
- (b) Does there exist a number which 7 greater than its cube?

[Solution:]

- (a) Consider the function $f(x) = \sqrt[7]{x} + x - 1$. We apply the I.V.T. to this function on the interval $[0, 1]$ with $N = 0$. The first thing we have to check is that $f(x)$ is continuous on $[0, 1]$. This is true because

- (i) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (\sqrt[7]{x} + x - 1) = -1 = f(0)$.
- (ii) $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (\sqrt[7]{x} + x - 1) = 1 = f(1)$.
- (iii) $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} (\sqrt[7]{x} + x - 1) = \sqrt[7]{a} + a - 1 = f(a)$ for all $a \in (0, 1)$.

So because $f(0) < 0 < f(1)$ we deduce from the I.V.T. that there exists a c where $f(c) = 0$. This c will solve the given equation.

- (b) In other words: is there a $c \in \mathbb{R}$ such that $c = c^3 + 7$? It is enough to prove that there exists a $c \in \mathbb{R}$ such that the function $f(x) = x^3 - x + 7$ takes the value $f(c) = 0$. To deduce this we apply the I.V.T. to $f(x)$ on the interval $[-3, 0]$. Note that:

- (i) $f(x)$ is a polynomial, so it is continuous at every point.
- (ii) $f(-3) = -17$ and $f(0) = 7$.
- (iii) $f(-3) < 0 < f(0)$.

So we can deduce from the I.V.T. that such a c will indeed exist.

Problem 4 Consider the function

$$f(x) = x^3 + 10 \sin x$$

Prove that there exists a real number c such that $f(c) = 800$.

[Solution:] We apply the I.V.T. (Intermediate Value Theorem) to this function on the interval $[0, 10]$. Note that:

- (1) The function is continuous at every $a \in \mathbb{R}$ because it is a combination of functions x and $\sin x$ that are known to be continuous everywhere.
- (2) $f(0) = 0$ and $f(10) = 10^3 + 10 \sin 10 \geq 10^3 - 10 = 990$.
- (3) $f(0) < 800 < 990 \leq f(10)$.

So the assumptions of the I.V.T. are satisfied on $[0, 10]$. We conclude that there exists a $c \in [0, 10]$ such that $f(c) = 800$.

Problem 5 Let $f(x) = \sqrt{16 - x}$. Use the definition of the derivative to determine its derivative function. State the domain of the function and the domain of the derivative.

[Solution:] The derivative of $f(x)$ at a point x is given by the following limit, if it exists:

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sqrt{16 - (x+h)} - \sqrt{16 - x}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sqrt{16 - (x+h)} - \sqrt{16 - x}}{h} \cdot \frac{\sqrt{16 - (x+h)} + \sqrt{16 - x}}{\sqrt{16 - (x+h)} + \sqrt{16 - x}} \\
 &= \lim_{h \rightarrow 0} \frac{[16 - (x+h)] - (16 - x)}{h [\sqrt{16 - (x+h)} + \sqrt{16 - x}]} \\
 &= \lim_{h \rightarrow 0} \frac{-h}{h [\sqrt{16 - (x+h)} + \sqrt{16 - x}]} \\
 &= \lim_{h \rightarrow 0} \frac{-1}{\sqrt{16 - (x+h)} + \sqrt{16 - x}} \\
 &= -\frac{1}{2\sqrt{16 - x}}.
 \end{aligned}$$

The domain of $f(x)$ is $(-\infty, 16]$. The domain of $f'(x)$ is $(-\infty, 16)$.

Problem 6 Let $g(x) = \frac{x^2-1}{2x-1}$. Use the definition of the derivative to determine its derivative function. State the domain of the function and the domain of the derivative.

[Solution:] The derivative of $g(x)$ at a point x is given by the following limit, if it exists:

$$\begin{aligned}
 g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{(x+h)^2-1}{2(x+h)-1} - \frac{x^2-1}{2x-1}}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \cdot \left(\frac{(x+h)^2-1}{2(x+h)-1} - \frac{x^2-1}{2x-1} \right) \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \left(\frac{[(x+h)^2-1](2x-1)}{[2(x+h)-1](2x-1)} - \frac{(x^2-1)[2(x+h)-1]}{(2x-1)[2(x+h)-1]} \right) \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{[(x+h)^2-1](2x-1) - (x^2-1)[2(x+h)-1]}{[2(x+h)-1](2x-1)} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{(x^2+2xh+h^2-1)(2x-1) - (x^2-1)(2x+2h-1)}{[2(x+h)-1](2x-1)} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{2x^2h+2xh^2-2xh-h^2+2h}{[2(x+h)-1](2x-1)} = \lim_{h \rightarrow 0} \frac{2x^2+2xh-2x-h+2}{[2(x+h)-1](2x-1)} \\
 &= \frac{2x^2-2x+2}{(2x-1)^2}.
 \end{aligned}$$

The domain of $g(x)$ is $\mathbb{R} \setminus \{\frac{1}{2}\}$. The domain of $f'(x)$ is $\mathbb{R} \setminus \{\frac{1}{2}\}$ as well.

Problem 7 Consider the function $f(x) = x^{\frac{1}{3}}$. Show that

- (a) Show that $f(x)$ is not differentiable at $x = 0$.
- (b) Calculate $f'(a)$ in the case that $a \neq 0$.
- (c) Show that the curve $y = x^{\frac{1}{3}}$ has a vertical tangent line at $(0, 0)$.
- (d) Illustrate by graphing $f(x)$ and $f'(x)$.

[Solution:]

- (a) To show that $f(x)$ is not differentiable at $x = 0$ we have to show that the limit which defines the derivative at that point, namely

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

does not exist.

The calculation is:

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^{\frac{1}{3}} - 0}{h} = \lim_{h \rightarrow 0} \frac{1}{h^{\frac{2}{3}}} = \infty.$$

So the limit does not exist. Hence $f(x)$ is not differentiable at $x = 0$.

(b) Let $a \neq 0$. Then

$$\begin{aligned}
 f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{(a+h)^{\frac{1}{3}} - a^{\frac{1}{3}}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(a+h)^{\frac{1}{3}} - a^{\frac{1}{3}}}{h} \cdot \frac{(a+h)^{\frac{2}{3}} + (a+h)^{\frac{1}{3}}a^{\frac{1}{3}} + a^{\frac{2}{3}}}{(a+h)^{\frac{2}{3}} + (a+h)^{\frac{1}{3}}a^{\frac{1}{3}} + a^{\frac{2}{3}}} \\
 &= \lim_{h \rightarrow 0} \frac{\left[(a+h)^{\frac{1}{3}}\right]^3 - \left(a^{\frac{1}{3}}\right)^3}{h \left[(a+h)^{\frac{2}{3}} + (a+h)^{\frac{1}{3}}a^{\frac{1}{3}} + a^{\frac{2}{3}}\right]} \\
 &= \lim_{h \rightarrow 0} \frac{(a+h) - a}{h \left[(a+h)^{\frac{2}{3}} + (a+h)^{\frac{1}{3}}a^{\frac{1}{3}} + a^{\frac{2}{3}}\right]} \\
 &= \lim_{h \rightarrow 0} \frac{1}{(a+h)^{\frac{2}{3}} + (a+h)^{\frac{1}{3}}a^{\frac{1}{3}} + a^{\frac{2}{3}}} \\
 &= \frac{1}{3a^{\frac{2}{3}}} = \frac{1}{3}a^{-\frac{2}{3}}.
 \end{aligned}$$

(c) To check that a graph has a vertical tangent line at a point, we have to check that it is

(i) continuous at that point, and that

(ii) $\lim_{x \rightarrow a} f'(x) = \infty$ or $-\infty$.

The given $f(x)$ satisfies:

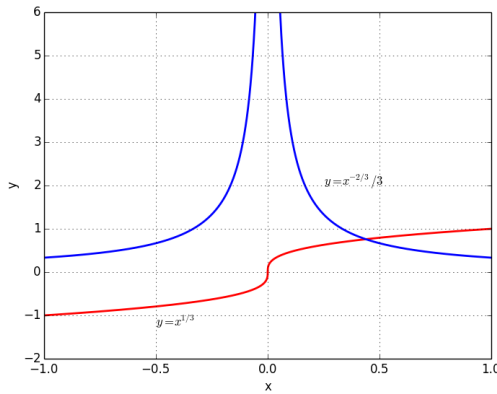
$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x^{\frac{1}{3}} = 0 = f(0).$$

So it is continuous at the point 0. Also, using the answer to part (b) we calculate

$$\lim_{x \rightarrow 0} f'(x) = \lim_{x \rightarrow 0} \frac{1}{3}x^{-\frac{2}{3}} = \infty.$$

So this graph has a vertical tangent at $x = 0$.

(d) The graphs of $f(x)$ and $f'(x)$ are shown in the below figure.



Problem 8 Using the definition of derivative, determine the derivative function of the function

$$f(x) = |x - 2|.$$

Sketch the graphs of $f(x)$ and of $f'(x)$.

[Solution:] This function is defined by different rules, depending on whether $x < 2$, $x = 2$ or $x > 2$. The rules are:

$$f(x) = \begin{cases} 2 - x, & \text{if } x < 2 \\ 0, & \text{if } x = 2 \\ x - 2, & \text{if } x > 2 \end{cases}$$

To calculate the derivative function, we are going to have to split this into 3 cases.

- Case 1: $a < 2$.

In this case, $f(x)$ is given by the rule $2 - x$ at every point close to a , so the calculation becomes:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{(2 - (a+h)) - (2 - a)}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1.$$

- Case 2: $a > 2$.

In this case, $f(x)$ is given by the rule $x - 2$ at every point close to a , so the calculation becomes:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{((a+h) - 2) - (a - 2)}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1.$$

- Case 3: $a = 2$.

The limit which determines the derivative at $a = 2$ is:

$$f'(2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$$

The function inside the limit is defined by two different rules, depending on whether $h > 0$ or $h < 0$, so to understand this limit we'll need to consider the corresponding 1-sided limits. The limit from the right is:

$$\lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0^+} \frac{((2+h) - 2) - 0}{h} = 1.$$

The limit from the left is:

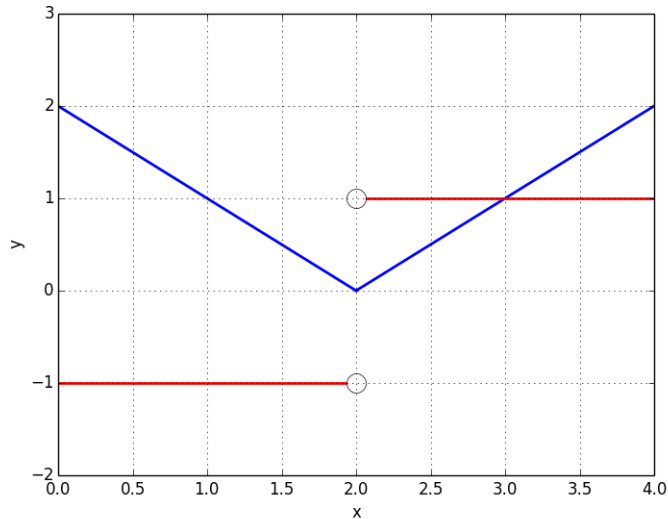
$$\lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0^-} \frac{(2 - (2+h)) - 0}{h} = -1.$$

The 1-sided limits are different, so the derivative is not defined at the point $x = 2$.

In summary,

$$f'(x) = \begin{cases} -1, & \text{if } x < 2 \\ \text{undefined}, & \text{if } x = 2 \\ 1, & \text{if } x > 2 \end{cases}$$

The graphs of $f(x)$ (in blue) and $f'(x)$ (in red) are:



Problem 9 Using the definition of derivative, determine the derivative function of the function

$$f(x) = |x^2 - 1|.$$

Sketch the graphs of $f(x)$ and of $f'(x)$.

[Solution:] This function is defined by different rules, depending on whether $x^2 < 1$, $x^2 = 1$ or $x^2 > 1$. The rules are:

$$f(x) = \begin{cases} x^2 - 1, & \text{if } x < -1 \\ 0, & \text{if } x = -1 \\ 1 - x^2, & \text{if } -1 < x < 1 \\ 0, & \text{if } x = 1 \\ x^2 - 1, & \text{if } x > 1 \end{cases}$$

To calculate the derivative function, we are going to have to split this into 4 cases.

- Case 1: $a < -1$ or $a > 1$.

In this case, $f(x)$ is given by the rule $x^2 - 1$ at every point close to a , so the calculation becomes:

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{[(a+h)^2 - 1] - (a^2 - 1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2ah + h^2}{h} = \lim_{h \rightarrow 0} (2a + h) \\ &= 2a. \end{aligned}$$

- Case 2: $-1 < a < 1$.

In this case, $f(x)$ is given by the rule $1 - x^2$ at every point close to a , so the calculation becomes:

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{[1 - (a+h)^2] - (1 - a^2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2ah - h^2}{h} = \lim_{h \rightarrow 0} (-2a - h) \\ &= -2a. \end{aligned}$$

- Case 3: $a = -1$.

The limit which determines the derivative at $a = -1$ is:

$$f'(-1) = \lim_{h \rightarrow 0} \frac{f(-1+h) - f(-1)}{h}$$

The function inside the limit is defined by two different rules, depending on whether $h > 0$ or $h < 0$, so to understand this limit we'll need to consider the corresponding 1-sided limits. The limit from the right is:

$$\lim_{h \rightarrow 0^+} \frac{f(-1+h) - f(-1)}{h} = \lim_{h \rightarrow 0^+} \frac{(1 - (-1+h)^2) - 0}{h} = \lim_{h \rightarrow 0^+} (2 - h) = 2.$$

The limit from the left is:

$$\lim_{h \rightarrow 0^-} \frac{f(-1+h) - f(-1)}{h} = \lim_{h \rightarrow 0^-} \frac{((-1+h)^2 - 1) - 0}{h} = \lim_{h \rightarrow 0^-} (h - 2) = -2.$$

The 1-sided limits are different, so the derivative is not defined at the point $x = -1$.

- Case 4: $a = 1$.

The limit which determines the derivative at $a = 1$ is:

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

The function inside the limit is defined by two different rules, depending on whether $h > 0$ or $h < 0$, so to understand this limit we'll need to consider the corresponding 1-sided limits. The limit from the right is:

$$\lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^+} \frac{((1+h)^2 - 1) - 0}{h} = \lim_{h \rightarrow 0^+} (h + 2) = 2.$$

The limit from the left is:

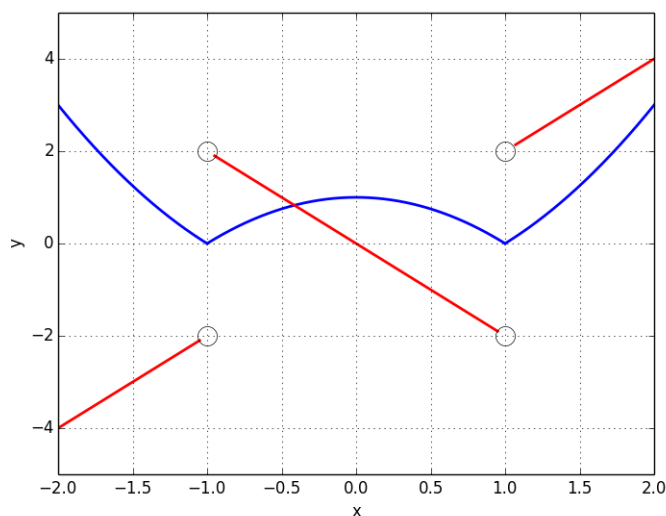
$$\lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^-} \frac{(1 - (1+h)^2) - 0}{h} = \lim_{h \rightarrow 0^-} (-h - 2) = -2.$$

The 1-sided limits are different, so the derivative is not defined at the point $x = 1$.

In summary,

$$f'(x) = \begin{cases} 2x, & \text{if } x < -1 \\ \text{undefined}, & \text{if } x = -1 \\ -2x, & \text{if } -1 < x < 1 \\ \text{undefined} & \text{if } x = 1 \\ 2x, & \text{if } x > 1 \end{cases}$$

The graphs of $f(x)$ (in blue) and $f'(x)$ (in red) are:



Problem 10 Consider the function $f(x) = x|x|$.

- (i) Graph this function.
- (ii) For what values of x is $f(x)$ differentiable?
- (iii) Find a formula for $f'(x)$.

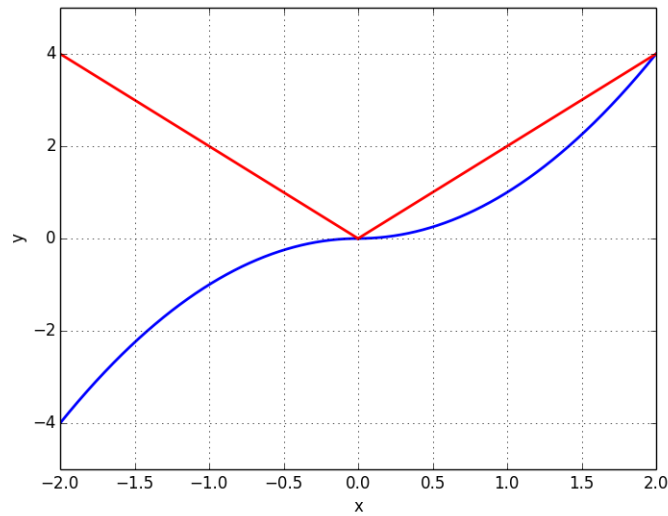
[Solution:] This function is determined by different rules in different regions:

$$f(x) = \begin{cases} -x^2, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \\ x^2, & \text{if } x > 0 \end{cases}$$

The graph of f is straightforward to assemble using the standard quadratic graph (blue curve below).

This graph appears to have a non-vertical tangent line at every point, so we expect that the function will be differentiable at every point. To answer with certainty, we'll apply the definition of derivative to calculate it.

To calculate the derivative function, we are going to have to split this into 3 cases.



- Case 1: $a < 0$.

In this case, $f(x)$ is given by the rule $-x^2$ at every point close to a , so the calculation becomes:

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{[-(a+h)^2] - (-a^2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2ah - h^2}{h} = \lim_{h \rightarrow 0} (-2a - h) \\ &= -2a. \end{aligned}$$

- Case 2: $a > 0$.

In this case, $f(x)$ is given by the rule x^2 at every point close to a , so the calculation becomes:

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{(a+h)^2 - a^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2ah + h^2}{h} = \lim_{h \rightarrow 0} (2a + h) \\ &= 2a. \end{aligned}$$

- Case 3: $a = 0$.

The limit which determines the derivative at $a = 0$ is:

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

The function inside the limit is defined by two different rules, depending on whether $h > 0$ or $h < 0$, so to understand this limit we'll need to consider the corresponding 1-sided limits. The limit from the right is:

$$\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{(0+h)^2 - 0}{h} = \lim_{h \rightarrow 0^+} h = 0.$$

The limit from the left is:

$$\lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{-(0+h)^2 - 0}{h} = \lim_{h \rightarrow 0^-} (-h) = 0.$$

The 1-sided limits are equal, so the derivative is defined at the point $x = 0$ and equals 0.

In summary, the function $f(x)$ is differentiable at every point, with the derivative given by

$$f'(x) = \begin{cases} -2x, & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 2x, & \text{if } x > 0 \end{cases}$$

Alternatively, $f'(x)$ can be written as $f'(x) = 2|x|$. The graph of $f'(x)$ is the red curve above.

Problem 11 The **left-hand derivative** and the **right-hand derivative** of a function $f(x)$ at a point a are defined by

$$f'_-(a) = \lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h} \quad \text{and} \quad f'_+(a) = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}$$

if these limits exist.

- (i) Briefly explain why f is differentiable at a if and only if both of these 1-sided derivatives exist and are equal.

- (ii) Now consider the function

$$f(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ 4 - x, & \text{if } 0 < x < 3 \\ \frac{1}{4-x}, & \text{if } x \geq 3 \end{cases}$$

Determine $f'_-(3)$ and $f'_+(3)$.

- (iii) Sketch the graph of $f(x)$.

- (iv) Where is f continuous?

- (v) Where does f fail to be differentiable?

[Solution:]

- (i) According to the definition, a function f is differentiable at a point a if and only if the following limit exists

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

And according to a standard theorem about limits, this limit exists if and only if the two corresponding 1-sided limits

$$\lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}$$

exist and are equal. These 2 limits are the “1-sided derivatives” introduced above.

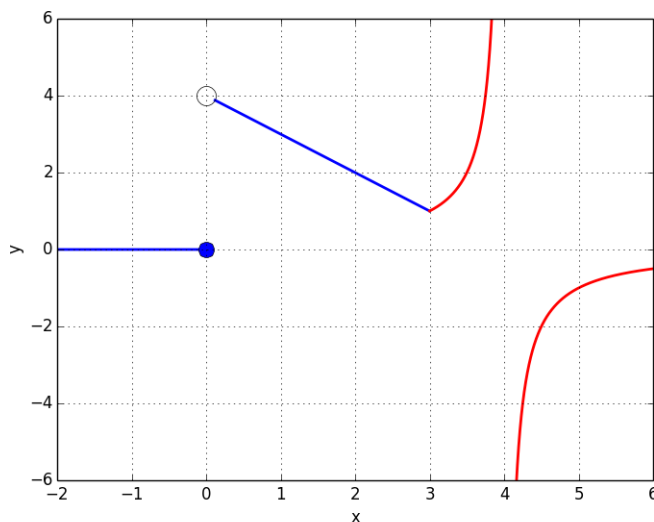
(ii) The left-hand derivative of $f(x)$ at the point $x = 3$ is given by the limit:

$$\begin{aligned} f'_-(3) &= \lim_{h \rightarrow 0^-} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0^-} \frac{[4 - (3+h)] - \frac{1}{4-3}}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{-h}{h} = \lim_{h \rightarrow 0^-} (-1) \\ &= -1. \end{aligned}$$

And the right-hand derivative of $f(x)$ at the point $x = 3$ is given by the limit:

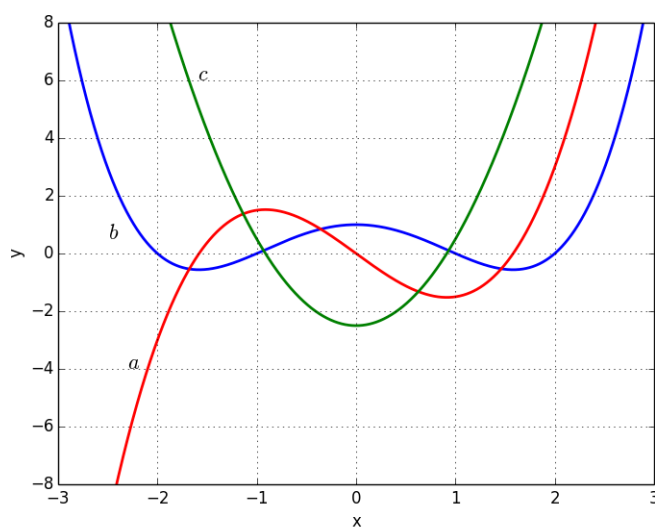
$$\begin{aligned} f'_+(3) &= \lim_{h \rightarrow 0^+} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0^+} \frac{\frac{1}{4-(3+h)} - \frac{1}{4-3}}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{\frac{h}{4-(3+h)}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{1-h} \\ &= 1. \end{aligned}$$

(iii) The graph is on the next page.



- (iv) The function is discontinuous at 0 (where it has a jump discontinuity) and at 4 (where it has an infinite discontinuity).
- (v) The function fails to be differentiable at 0 and 4 (because it fails to even be continuous there) and also fails to be differentiable at $x = 3$, where it has a corner. (To be precise, at $x = 3$ the left and right derivatives exist but are unequal.)

Problem 12 Given a certain function $f(x)$, below we have plotted $f(x)$, $f'(x)$, and $f''(x)$. Match these functions to the different graphs a , b and c . Justify your answer.



[Solution:] The way to do this is to find the points where the graphs have horizontal tangent lines. We know at those points the derivatives are 0. At $x = 0$, both b and c are flat. So their derivatives at $x = 0$ should be 0. Graph a is 0 at $x = 0$. So Graph a must be the derivative of b or of c . If Graph a is the derivative of c , and c is decreasing to the left of $x = 0$, Graph a should be below the x -axis when $x < 0$. But this is not true. So Graph a is the derivative of Graph b . If Graph b is the derivative of Graph c , Graph b should be below the x -axis when $x < 0$. Again this is not true. So the only possibility now is that Graph c is the derivative of Graph a .

Thus: Graph b depicts $f(x)$, Graph a depicts $f'(x)$, and Graph c depicts $f''(x)$.